SIMATS ENGINEERING

CSA6303- Threat Intelligence and Network Security

A1-Threat Intelligence Report

Assessment Questions

Subject	Threat Intelligence	**Total Marks**	50
Questions	5 (all compulsory)	**Marks per Question**	10
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Instructions: Answer all questions. Each question carries 5 marks. Support answers with examples and diagrams where appropriate.

CO3: Perform threat analysis using intelligence models and utilize industry tools for monitoring and detection.

Code of Conduct:

I N.Harshith (192311129) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

N.Harshith

Signature of Student

Questions :

1. Threat Hunting Techniques

Threat hunting is a **proactive cybersecurity activity** used to identify hidden threats that may bypass traditional security controls.

Investigation Approach

Step 1: Establish a Hypothesis

- Suspect unauthorized network access due to unusual login times, unknown processes, and external connections.

Step 2: Analyze Authentication Logs

- Review user login records.
- Identify:
    - Failed login attempts
    - Logins outside business hours
    - Logins from unusual locations
    - Privilege escalation events

Step 3: Examine Endpoint Activity

- Investigate unfamiliar processes.
- Check process names, parent-child relationships, and digital signatures.
- Look for malware persistence mechanisms such as:
    - Startup entries
    - Scheduled tasks
    - Services

Step 4: Monitor Network Connections

- Identify communication with unfamiliar IP addresses and domains.
- Analyze outbound traffic patterns.
- Detect command-and-control (C2) communications.

Step 5: Collect Indicators of Compromise (IoCs)

- File hashes
- Suspicious domains
- IP addresses

- Registry keys
- Process names

Step 6: Correlate Events

- Match login events with process execution and network activity.
- Determine whether the activities are related.

Outcome

Threat hunting helps uncover hidden attackers, identify compromised hosts, and collect evidence for incident response.

2. SIEM Tools and Security Monitoring

A **Security Information and Event Management (SIEM)** system centralizes log collection, monitoring, and alerting.

Event Collection

SIEM collects logs from:

- Servers
- Firewalls
- Endpoints
- Applications
- Network devices

Event Analysis

The SIEM normalizes and analyzes data to detect:

- Brute-force login attempts
- Unauthorized access
- Malware activity
- Suspicious network communications

Alert Correlation

Instead of treating alerts separately, the SIEM correlates them.

Example

1. Multiple failed logins detected.
2. Successful login occurs.
3. Suspicious process launches.
4. External communication begins.

The SIEM links these events into a single incident.

Prioritization

Security analysts prioritize threats using:

- Severity ratings
- Number of affected systems
- Asset criticality
- Threat intelligence matches

Benefits

- Real-time monitoring
- Faster threat detection
- Centralized visibility
- Reduced analyst workload

Conclusion

SIEM systems improve security monitoring by correlating events and helping analysts respond to high-risk incidents efficiently.

3. Security Monitoring and Anomaly Detection

Scenario

A database server suddenly generates large outbound traffic during non-working hours.

Investigation Steps
Step 1: Verify the Anomaly

- Compare current traffic with historical baselines.
- Confirm the activity is unusual.

Step 2: Analyze Network Logs

- Identify destination IP addresses.
- Determine traffic volume and protocols used.

Step 3: Review Server Logs

- Examine:
    - User logins
    - Administrative actions
    - Database queries
    - File access records

Step 4: Look for Data Exfiltration Indicators

Signs include:

- Large file transfers
- Compressed archives
- Encrypted outbound traffic
- Connections to unknown external hosts

Step 5: Determine Root Cause

Possible causes:

- Malware infection
- Insider threat
- Compromised credentials
- Misconfigured applications

Prevention Measures

- Enable Data Loss Prevention (DLP)
- Network segmentation
- Continuous monitoring
- User activity monitoring
- Multi-factor Authentication (MFA)
- Outbound traffic restrictions

Outcome

Timely investigation can detect data exfiltration and prevent data breaches.

4. Tools Used in the IT Industry for Threat Hunting

When malware is suspected on multiple workstations, several security tools should be used together.

1. SIEM Tools

Examples: Splunk, Microsoft Sentinel, QRadar

Purpose:

- Collect logs
- Correlate events
- Detect suspicious activities

Benefits:

- Centralized investigation
- Attack timeline creation

2. Endpoint Detection and Response (EDR)

Examples: CrowdStrike Falcon, Microsoft Defender for Endpoint, SentinelOne

Purpose:

- Monitor endpoints
- Detect malicious processes
- Isolate infected systems

Benefits:

- Real-time visibility
- Rapid containment

3. Network Traffic Analyzers

Examples: Wireshark, Zeek, TCPDump

Purpose:

- Analyze network packets
- Detect malicious communications
- Identify command-and-control traffic

Benefits:

- Reveals attacker communication channels

4. Vulnerability Scanners

Examples: Nessus, OpenVAS, Qualys

Purpose:

- Discover security weaknesses
- Identify unpatched systems

Benefits:

- Finds exploitation opportunities used by attackers

Recommended Approach

1. Use SIEM for log correlation.
2. Use EDR to identify infected machines.
3. Use network analyzers to inspect communication.
4. Use vulnerability scanners to find exploited weaknesses.
5. Isolate affected systems and remediate threats.

5. Integrated Threat Hunting and Incident Reporting

Step 1: Data Collection

Gather logs from:

- Endpoints
- Firewalls
- DNS servers
- Authentication systems
- Web servers
- Proxy servers

Using:

- Splunk
- ELK Stack (Elasticsearch, Logstash, Kibana)

Step 2: Log Correlation

Correlate:

Event Type	Observation
Login Logs	Suspicious logins
DNS Logs	Malicious domain lookups
Endpoint Logs	Unknown process execution
Firewall Logs	External connections

Build a timeline of attacker activities.

Step 3: Threat Investigation

Investigate:

- Unknown user logins
- Process creation events
- New services
- Registry modifications
- Privilege escalation attempts

Identify affected systems and accounts.

Step 4: Threat Intelligence Validation

Check IoCs using:

- VirusTotal
- MalwareBazaar
- AlienVault OTX
- IBM X-Force Exchange

Verify:

- File hashes
- Domains
- IP addresses
- Malware family associations

Step 5: Malware Analysis (If Required)

Static Analysis

- Hash analysis
- String extraction
- Import analysis

Dynamic Analysis

- Sandbox execution
- Process monitoring
- Network monitoring

Determine malware capabilities and behavior.

Step 6: Risk Assessment

Assess impact on:

Area	Risk
Confidentiality	High
Integrity	High
Availability	Medium

Determine:

- Scope of compromise
- Critical assets affected
- Business impact

Step 7: Remediation Actions

Containment

- Isolate infected devices
- Block malicious domains and IPs

- Disable compromised accounts

Eradication

- Remove malware
- Patch vulnerabilities
- Eliminate persistence mechanisms

Recovery

- Restore systems from clean backups
- Reset passwords
- Verify system integrity

Step 8: Threat Hunting Report
Report Structure

1. Executive Summary
2. Incident Description
3. Data Sources Used
4. Investigation Methodology
5. Timeline of Events
6. Malware Analysis Findings
7. Indicators of Compromise (IoCs)
8. Threat Intelligence Results
9. Risk Assessment
10. Containment Actions
11. Recovery Actions
12. Recommendations
13. Conclusion

Conclusion

The integrated threat-hunting process combines log collection, SIEM/ELK analysis, IoC validation, malware analysis, and risk assessment to detect threats quickly, minimize damage, and strengthen the organization's security posture.